

scPTR: Decomposing Post-Transcriptional Regulation at Single-Cell Resolution

Introduction

Single-cell RNA sequencing jointly captures spliced and unspliced transcripts, yet computational frameworks have focused almost exclusively on transcriptional dynamics—most notably RNA velocity, which models the rate of change of spliced mRNA abundance [1,2]. Meanwhile, mRNA degradation—orchestrated by RNA-binding proteins (RBPs), microRNAs, and *cis*-regulatory elements—is a major determinant of steady-state gene expression and is increasingly recognized as a driver of cell fate transitions [3]. Despite this, no general framework exists to systematically estimate per-cell mRNA degradation rates, discover cell populations defined by post-transcriptional programs, or infer the regulatory networks governing mRNA stability at single-cell resolution. We present **scPTR** (Single-Cell Post-Transcriptional Regulatory Decomposition), a framework that estimates per-cell, per-gene mRNA degradation rates (γ) from standard scRNA-seq data and treats γ as a primary analytical axis—complementary to, and independent of, RNA velocity.

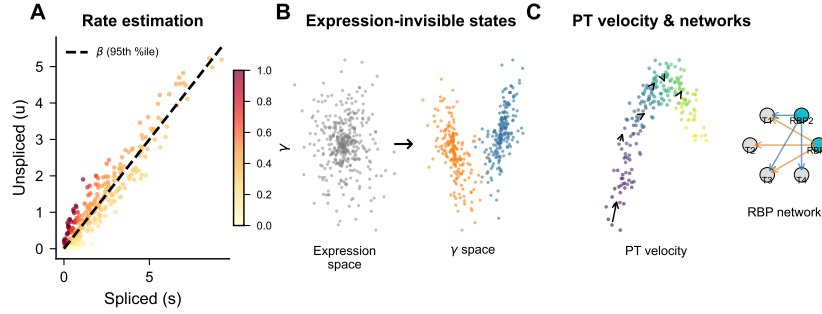


Figure 1: scPTR framework. (A) Splicing rates (β) estimated via quantile regression on phase portraits; per-cell $\gamma = \beta \cdot u/s$ after k -NN Gaussian smoothing. (B) Clustering in γ -space reveals post-transcriptional states invisible to expression analysis. (C) PT velocity (neighbor-averaged γ gradient) and library-size-corrected elastic net regression infer RBP–target networks.

Methods

For each gene, scPTR estimates the splicing rate β as the 95th-percentile slope of the unspliced/spliced phase portrait, then computes per-cell degradation rates via the kinetic steady-state relation $\gamma_{ig} = \beta_g \cdot u_{ig}/s_{ig}$, where u and s are k -NN Gaussian-kernel-smoothed counts. PT states are discovered by PCA, k -NN graph construction, and Leiden clustering in γ -space. PT velocity is the weighted mean γ difference between each cell and its neighbors, yielding a vector field orthogonal to RNA velocity. RBP–target networks are inferred via elastic net regression of target γ on smoothed RBP expression, with library-size partial correlation to correct for destabilizing bias. A structured VAE (DeepPTR) with a kinetic decoder disentangles transcriptional (z_T) and post-transcriptional (z_{PT}) latent spaces under a negative binomial observation model.

Results

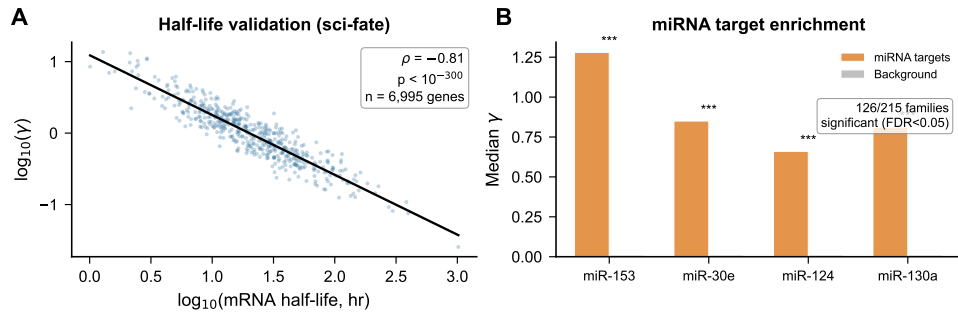


Figure 2: Validation of γ estimates. (A) Per-gene median γ correlates with published mRNA half-lives (Spearman $\rho = -0.81$, sci-fate; -0.35 to -0.40 in 10x data). (B) miRNA targets show elevated γ vs. background (126/215 families significant at FDR < 0.05; aggregate $p = 4.7 \times 10^{-65}$).

Gamma estimates were validated against published mRNA half-lives across three species–platform combinations, achieving Spearman correlations from -0.35 (10x mouse developmental data) to -0.81 (sci-fate human metabolic labeling; Fig. 2A). Predicted miRNA targets from TargetScan 8.0 were enriched among high- γ genes for 59% of 215 tested families (Fig. 2B), and γ correlated positively with 3' UTR length ($\rho = 0.34$, $p < 10^{-200}$).

Clustering in γ -space revealed expression-invisible PT states in 3/8 pancreatic and 6/11 hippocampal cell types—

subpopulations with distinct degradation programs yielding higher silhouette scores in γ -space than expression space (Fig. 3A). These were enriched for coherent pathways: ER protein processing in pancreas; synaptic plasticity in dentate gyrus.

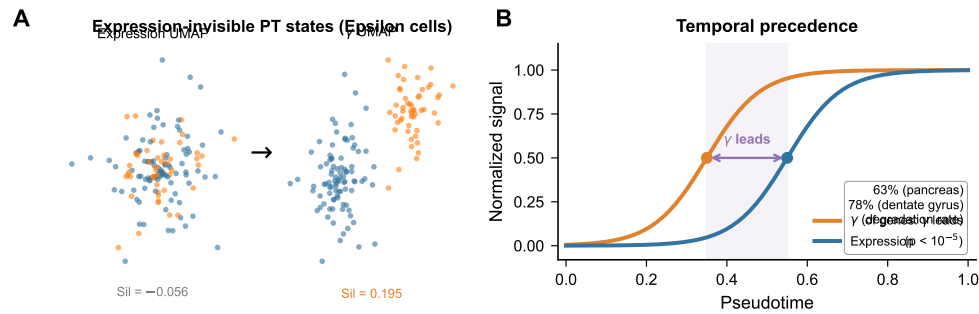


Figure 3: Key findings. (A) Pancreas Epsilon cells form one cluster in expression space (silhouette = -0.056) but separate into distinct PT states in γ -space (silhouette = 0.195). (B) Along pseudotime, γ changes precede expression changes for 63–78% of transition genes ($p < 10^{-5}$).

PT velocity was orthogonal to RNA velocity (cosine similarity ≈ 0) and degradation-rate changes preceded expression changes for 63–78% of transition genes (Fig. 3B). Library-size-corrected RBP–target network inference identified hub regulators (HNRNPA1, YBX1, ELAVL1) validated by DepMap CRISPR essentiality ($p = 6.4 \times 10^{-5}$). Application to neuroblastoma revealed a predominantly stabilizing RBP landscape (66% stabilizing edges), contrasting with developmental tissues.

Conclusions

scPTR reveals a previously inaccessible layer of post-transcriptional regulation in single-cell data. By treating mRNA degradation rates as per-cell observables, scPTR discovers expression-invisible cell states, shows that post-transcriptional rewiring precedes transcriptional changes during fate transitions, and identifies essential RBP regulators—establishing degradation-rate analysis as a complement to RNA velocity.

Availability: <https://github.com/bryanc5864/scPTR>

References

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